DO NOW — RECALL

Computational problem recognition

Name: _____ Class: _____ Date: _____

1. Why may Spiral be preferable for an uncertain high-risk project?

2. What is abstraction?

3. What is a base case?

4. For C# array int[] x={2,4,6,8}, a loop doubles elements at even indexes (0,2). Give the final array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It prioritises identifying and addressing risks in each cycle.

2. Model answer

Selecting relevant detail and omitting irrelevant detail.

3. Model answer

A condition that ends recursion.

4. Model answer

Final [4,4,12,8]. Sum=28. Indexes 0 and 2 changed; 1 and 3 unchanged.

Vocabulary — know these terms

problem recognition — Determine what problem is being solved and its computational form.

decomposition — Split into manageable subproblems.

divide and conquer — Divide input, solve smaller instances, combine results.

constraint — Condition a valid solution must satisfy.

tractability — Whether an algorithm can solve realistic sizes with feasible resources.

NEW LEARNING

Read, annotate and explain

Recognise ambiguity

'Fair' is subjective until objective criteria are supplied, such as balancing first-period lessons across teachers.

Make requirements testable

Represent rooms, lessons and slots; specify hard constraints and a success test.

Select techniques appropriately

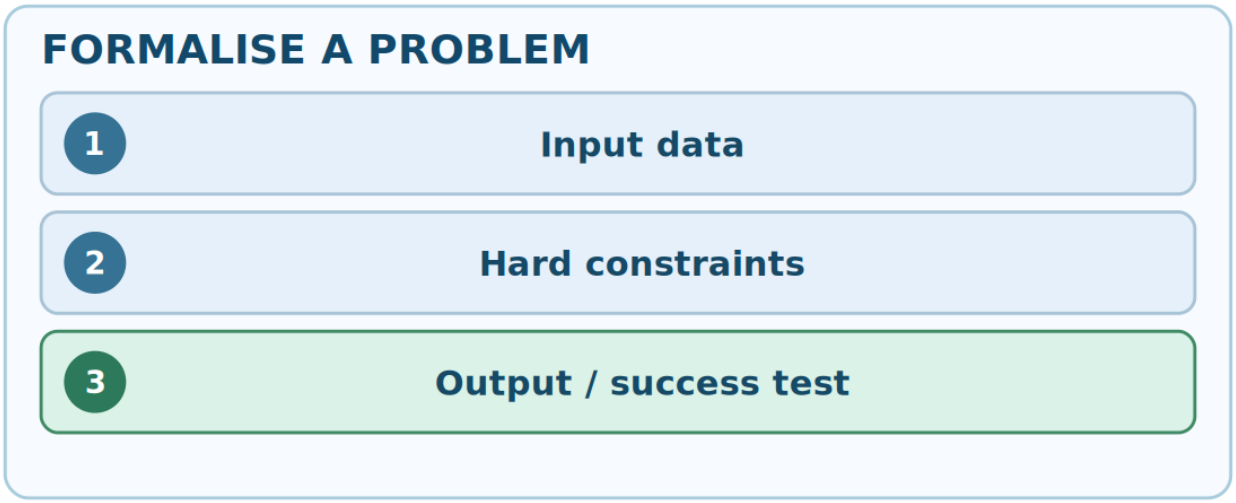
Decompose into assign/check/evaluate modules. Divide and conquer is useful where subproblems can be solved and combined consistently; not every decomposition is divide and conquer.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

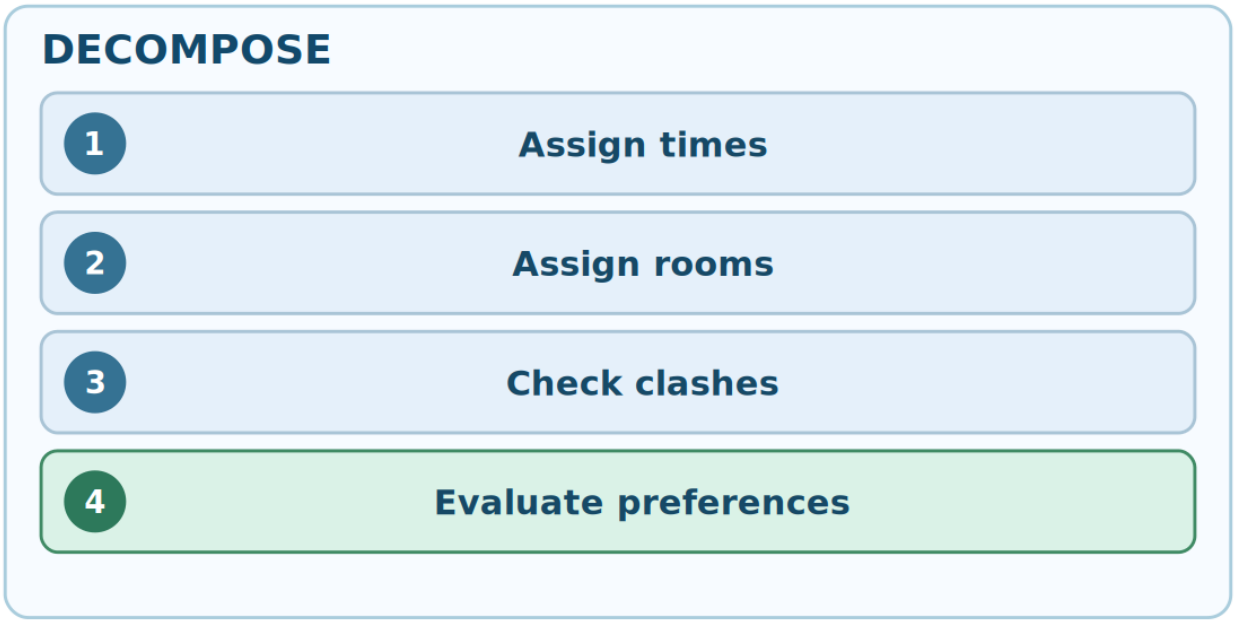
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



FORMALISE A PROBLEM; Input data; Hard constraints; Output / success test

New Learning

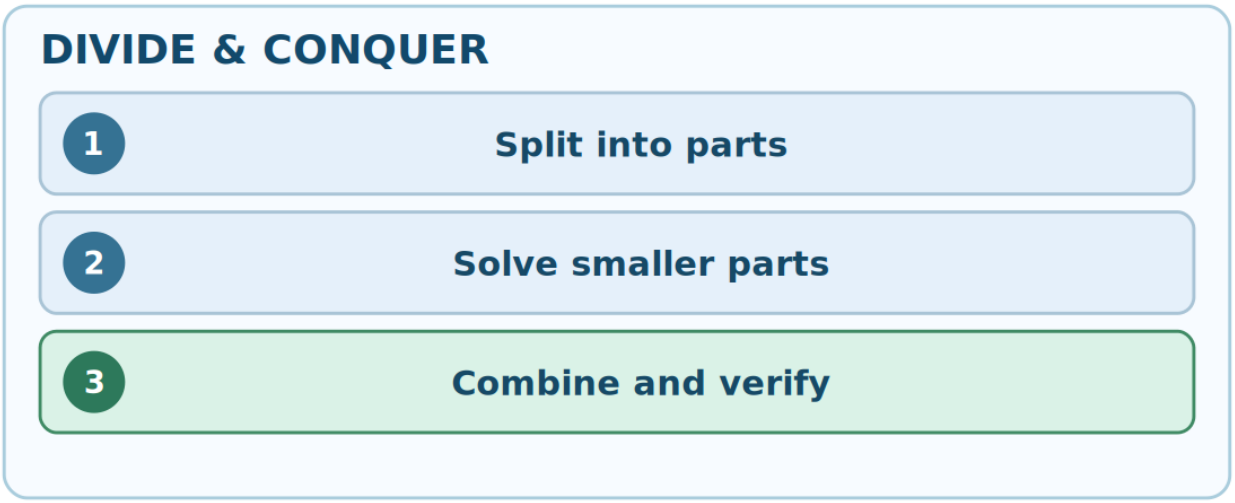


DECOMPOSE; Assign times; Assign rooms; Check clashes; Evaluate preferences

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



DIVIDE & CONQUER; Split into parts; Solve smaller parts; Combine and verify

I DO — WORKED EXAMPLE

Follow the worked method

Rewrite 'choose the best delivery van route' into a computational task.

Predict the method

Replace vague 'best' with a measurable objective such as minimum total travel time.

Check and explain

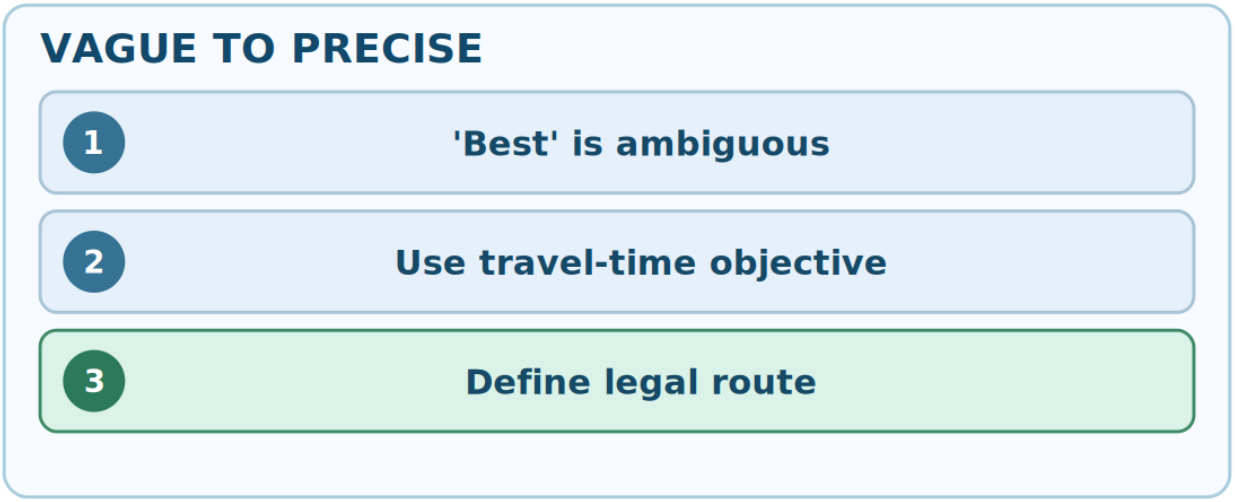
Specify route constraints and verify a produced route covers all required locations.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Rewrite 'choose the best delivery van route' into a computational task.

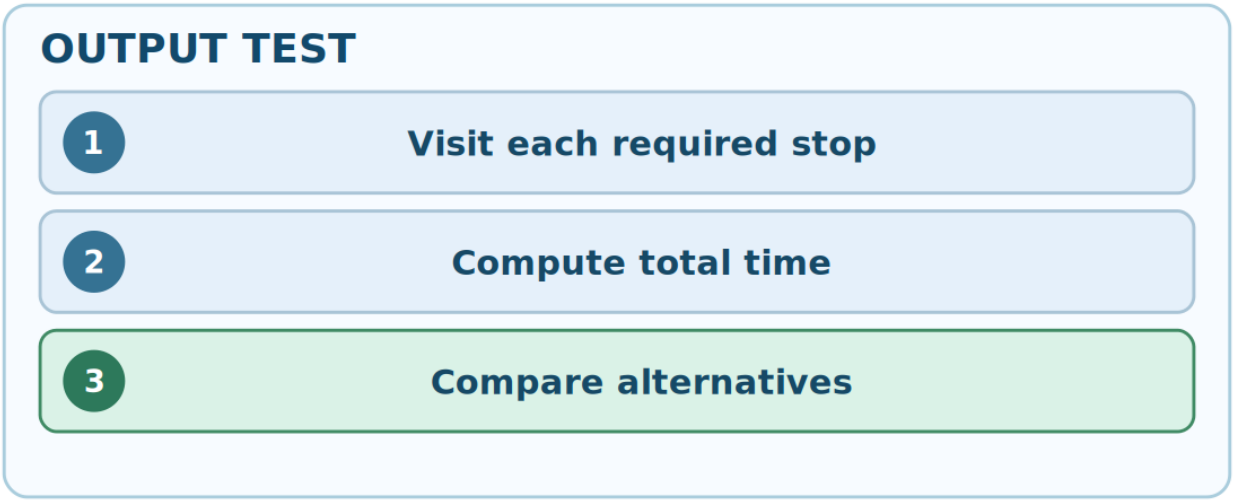
Replace vague 'best' with a measurable objective such as minimum total travel time.



VAGUE TO PRECISE; 'Best' is ambiguous; Use travel-time objective; Define legal route

Rewrite 'choose the best delivery van route' into a computational task.

Replace vague 'best' with a measurable objective such as minimum total travel time.



OUTPUT TEST; Visit each required stop; Compute total time; Compare alternatives

WE DO — GUIDED PRACTICE

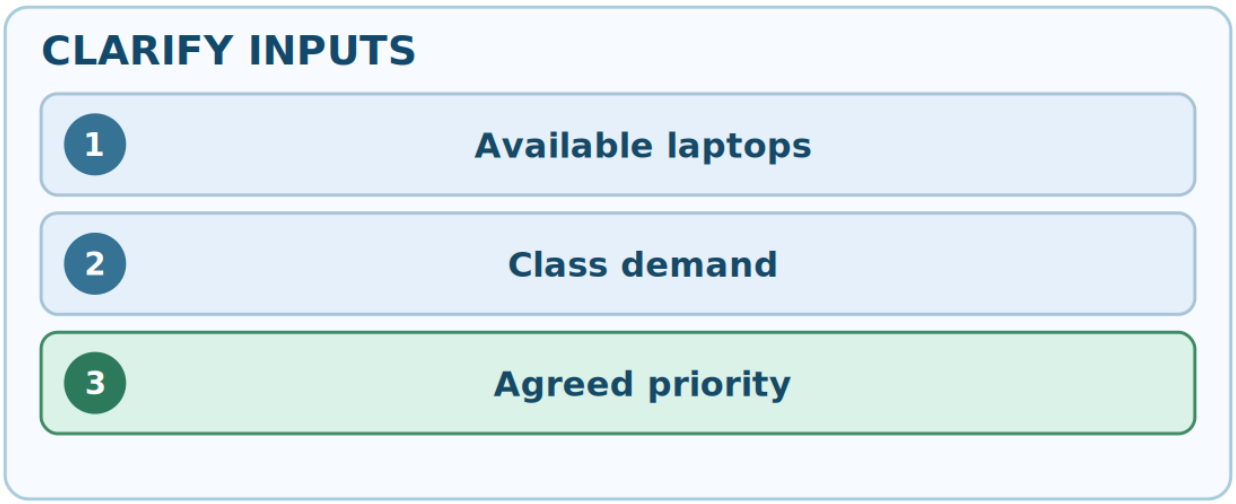
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants the 'fairest' allocation of laptops across classes. Make it computable.

Use explicit demand and stock data and define what fairness means with measurable priorities.



CLARIFY INPUTS; Available laptops; Class demand; Agreed priority

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the method Use explicit demand and stock data and define what fairness means with measurable priorities.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For a system allocating parking spaces, give one input, one constraint, one output and one test of correctness.

Question 2

Explain why 'select the nicest library books' is not computable as stated; propose precise inputs, a ranking rule and an output.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare decomposition and divide-and-conquer for sorting 1,000 items, using merge sort as an example.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An unclear human judgement becomes computable simply by choosing a faster processor.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

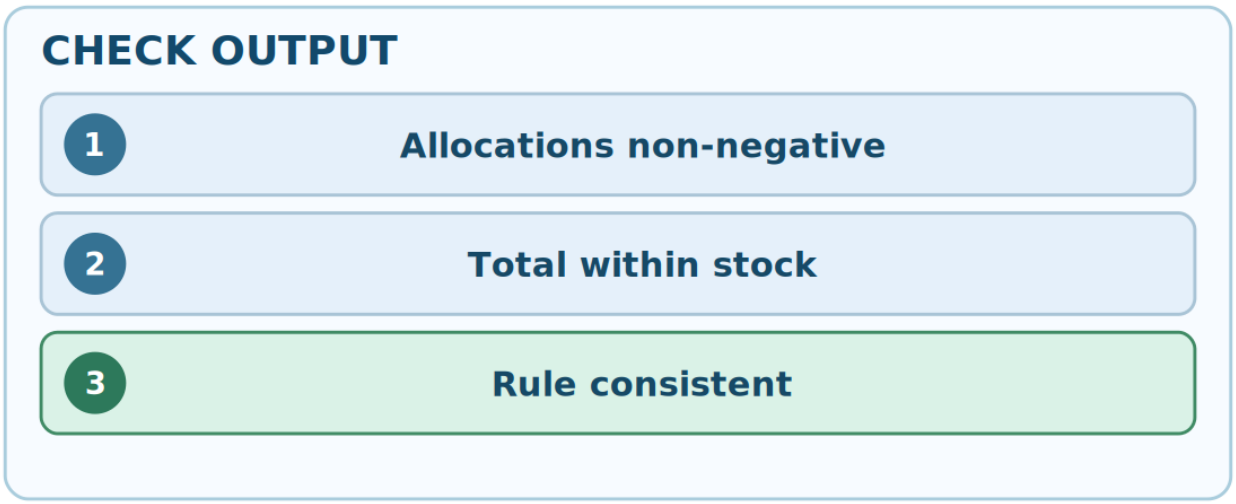
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants the 'fairest' allocation of laptops across classes. Make it computable.

Use explicit demand and stock data and define what fairness means with measurable priorities.



CHECK OUTPUT; Allocations non-negative; Total within stock; Rule consistent

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check and explain

Check total allocated never exceeds stock and each class allocation follows the stated policy.

IMPROVE: Explain one correction to your own reasoning.